

Camouflaged Human Detection

20201780 곽창민

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Introduction

제약조건

- Automatic First-Frame Detection
- Without Future-Frame
- Official Resources Required



Baseline

Model	MoCH IoU
EMIP	0.364
TSP-SAM	0.293
ZoomNeXt	0.389



Baseline : ZoomNeXt

Dataset : Evaluation



MoCH

Moving Camouflaged Human Object

CamLock

정답 mask



1차 mask



CamLock Module



추론 정책: onset 이전에는 넓은 범위에서 성능 방어, onset 이후에는 강하게 신뢰

Dataset : Training



MoCA

Moving Camouflaged Animals



CAD

Camouflaged Animal Dataset



CAMotion

Camouflaged Moving Object



초반 frame 증강해서 사용

Results

Model	MoCH IoU
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ZoomNeXt

0.3889

ZoomNeXt-CamLock

0.4602

Model	MoCH IoU
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ZoomNeXt (Before Motion)

0.2501

ZoomNeXt-CamLock (Before Motion)

0.2871

Model	MoCH IoU
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ZoomNeXt (After Motion)

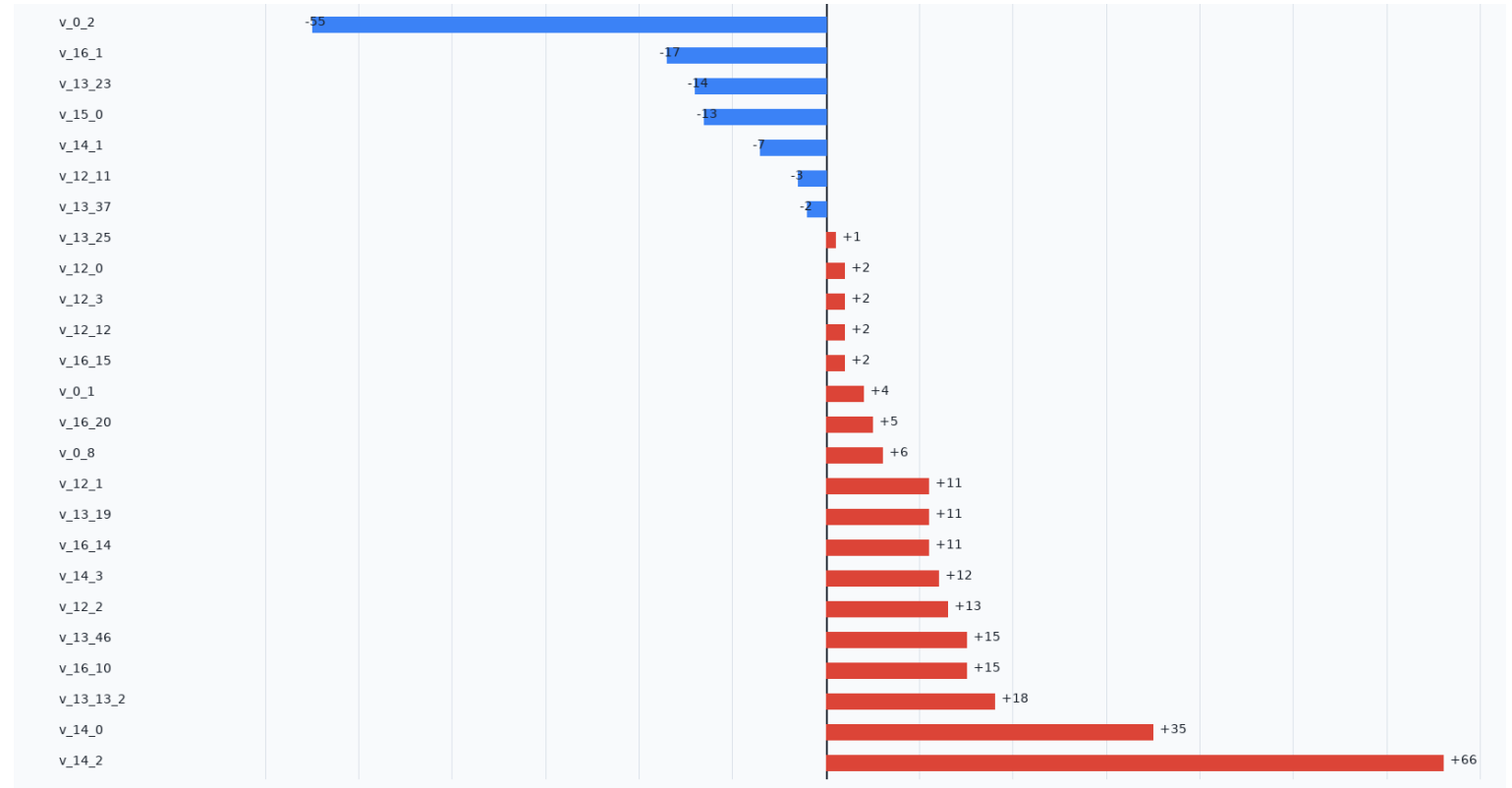
0.4745

ZoomNeXt-CamLock (After Motion)

0.5769

Limitations

Folder	실제 시작	CamLock onset	Diff
v_12_15	1	4	+3
v_12_3	2	4	+2
v_14_0	46	81	+35
v_14_1	75	68	-7
v_15_0	34	21	-13
v_16_1	28	11	-17
v_16_14	18	29	+11
v_16_15	12	14	+2
v_16_5	1	19	+18
v_0_1	2	6	+4
v_0_2	59	4	-55
v_0_8	2	8	+6
v_12_0	2	4	+2
v_12_11	17	14	-3
v_12_12	2	4	+2
v_12_2	23	36	+13
v_13_19	20	31	+11
v_13_23	23	9	-14
v_13_37	17	15	-2
v_13_46	16	31	+15
v_14_2	82	148	+66
v_14_3	2	14	+12
v_16_10	44	59	+15
v_12_1	2	13	+11
v_13_13_2	2	20	+18
v_13_25	8	9	+1
v_16_20	30	35	+5



개선 방안 : 1차 mask 성능 향상